

# IMDADULLAH RAJI

Mechanical Engineering, BUET · B.Sc. expected June 2027  
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## Research Interests

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Unsteady fluid dynamics · Dynamical systems · Data-driven modeling · Flow control

## Education

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**Bangladesh University of Engineering and Technology, Dhaka** 2022–2027 (Expected)  
*B.Sc. in Mechanical Engineering · CGPA: 3.58*

*Relevant coursework:* Complex Analysis, Differential Equations, Linear Algebra, Numerical Analysis, Computer Programming, Fluid Mechanics I–II

**Notre Dame College, Dhaka** 2019–2022  
*Higher Secondary Certificate, Science*

## Research Experience

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**Modeling and Control of Flows Over Rapidly Maneuvering Airfoils** 2026–present  
*Undergraduate Thesis · Supervisor: Dr. Mohammad Ali, Professor, Department of Mechanical Engineering, BUET*

- Assessing how accurately reduced-order models capture aerodynamic forces under active actuation, with leading-edge blowing as the control input, and developing improved formulations.
- Generating training data from controlled OpenFOAM simulations. The Reynolds number is kept low deliberately so the flow resolves fully without turbulence modeling, keeping the data faithful to the governing equations rather than to a closure model.

**Statistical Modeling of Lagrangian Coherent Structures** 2026–present

- Addressing computational bottlenecks in LCS calculations: integrating a full particle ensemble requires Jacobians and singular values at every step, and most of that cost goes to trajectories carrying no information about where attracting sets form.
- Investigating whether early integration steps predict where attracting sets form, allowing dense Lagrangian sampling to be concentrated in critical flow regions.

**Autonomous UAV–UGV Swarm Coordination** 2025–2026  
*Supervisor: Dr. Kazi Arafat Rahman, Associate Professor, Department of Mechanical Engineering, BUET*

- Implemented a collaborative, autonomous multi-agent UAV–UGV system using edge-computing hardware (Jetson Orin Nano, Raspberry Pi).

## Computational Projects

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**flowkit** — Python library under development for OpenFOAM post-processing, including snapshot loading, interpolation onto Cartesian grids, spatial cropping, and Lagrangian trajectory analysis.

**openfoam\_cases** — Reproducible CFD cases developed for the undergraduate thesis. Each documents its modeling assumptions and ships the geometry and meshing scripts needed to reproduce it.

## Publications

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N. A. Tanisha, **I. Raji**, A. Guha, N. Anita. *MuADDIBBS: A Hierarchical Outdoor Swarm System with Aerial Supervision Guiding Local Interaction*. 15th International Conference on Mechanical Engineering (ICME), BUET, 2025. [SSRN](#).

## Scientific Outreach & Honors

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### Bangladesh Olympiad on Astronomy and Astrophysics

*Academic Coordinator (2026–present); Academic Member (2021–present)*

- Set and evaluate problems for regional and national astrophysics olympiads; organize the national camp and the selection process for Bangladesh’s IOAA team.

**Honorable Mention**, International Olympiad on Astronomy and Astrophysics (IOAA), 2021

**Bronze Medal**, Open World Astronomy Olympiad (OWAO), 2022

## Technical Skills

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| <b>CFD</b>                  | OpenFOAM and gmsh; ANSYS Fluent                |
| <b>Scientific Computing</b> | Python (NumPy, SciPy); FEniCS; MATLAB/Simulink |
| <b>Data &amp; Modeling</b>  | PySINDy, PyDMD, scikit-learn; PyTorch          |
| <b>Development</b>          | Linux, Git, L <sup>A</sup> T <sub>E</sub> X    |